

Smita Sahay-Kausar

The University of Toledo College of Medicine & Life Sciences (UTCOMLS)
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EDUCATION

Doctor of Medicine The University of Toledo College of Medicine & Life Sciences, <i>Toledo, OH</i>	Expected May 2027
Doctor of Philosophy, Neuroscience and Neurological Disorders The University of Toledo College of Medicine & Life Sciences, <i>Toledo, OH</i>	May 2025
Post Baccalaureate, Biomedical Science California State University, East Bay, <i>Hayward, CA</i>	Aug 2018
Bachelor of Science, Bioengineering: Bioinformatics University of California, San Diego, <i>La Jolla, CA</i>	Jun 2017
High School Diploma Skyline High School, <i>Salt Lake City, UT</i>	May 2012

LICENCES & CERTIFICATIONS

United States Medical License - Step II	Jul 2026
United States Medical License - Step I	Apr 2022
Epilepsy First Aid Certification	Sep 2022
Adult Vision Screener Certification	Sep 2022
Advanced Cardiovascular Life Support (ACLS) Provider Certification	Feb 2022
Basic Life Support (BLS) Provider Certification	Feb 2020

RESEARCH INTERESTS

My research centers on molecular signaling, metabolism, and translational bioinformatics. I am interested in how these pathways intersect with neuropsychiatric diseases, neural dysfunction, & sensory disorders with the goal of integrating computational and translational approaches into clinically meaningful research.

RESEARCH EXPERIENCE

Pre-Doctoral Research Assistant: Cognitive Disorders Research Lab <i>Mentors:</i> Dr. Robert E. Smith, M.D., Ph.D. & Dr. Sinead M. O'Donovan, Ph.D. <i>Department:</i> Neurosciences The University of Toledo College of Medicine & Life Sciences, Toledo, OH <i>Scope:</i> Investigating metabolic dysfunction, ketogenic interventions, altered kinase networks, purinergic system, & differential gene expression across neuropsychiatric illnesses such as schizophrenia (SCZ), bipolar disorder (BPD), major depressive disorder (MDD), & suicide	May 2022-May 2025
Graduate Research Assistant: Cognitive Disorders Research Lab <i>Mentors:</i> Dr. Robert E. Smith, M.D., Ph.D. & Dr. Sinead M. O'Donovan, Ph.D. <i>Department:</i> Neurosciences The University of Toledo College of Medicine & Life Sciences, Toledo, OH	May 2021-Jul 2021

Scope: Optimized protein activity assays to gain insight into kinase networks in SCZ; Investigated purinergic receptors & downstream inflammation in SCZ; Investigated pathophysiology of MDD

Undergraduate Research Assistant

Jan 2018-May 2019

Mentor: Dr. Winston Tseng, Ph.D.; *Department:* Pre-Professional Health Academic Program

California State University, East Bay, Hayward, CA

Scope: Analyzed bacterial data from Illumina Inc. sequencers to find structural variants that lead to antibiotic resistance in patients

Undergraduate Research Assistant | WorldCare Technologies

Jun 2016-Jun 2017

Mentor: Dr. David Smith, M.D.; *Departments:* Global Teams in Engineering Service (Global TIES) & Biomedical Engineering

University of California, San Diego, La Jolla, CA

Scope: Developed cost-effective prototype for an open viral load (OVL) HIV monitoring device for the underserved population in Tijuana, Mexico; Designed PCR microwell chips for the OVL-HIV device for proper blood dilution reactions; Supported launch of second-generation prototype / kicked off clinical validation process to begin field implementation globally

Undergraduate Research Assistant

Jan 2016-Jun 2016

Mentors: Dr. Joseph Gleeson, M.D. & Dr. Niema Moshiri, Ph.D.; *Department:* Neurogenetics

University of California, San Diego School of Medicine, La Jolla, CA

Scope: Designed ultra-sensitive pipelines to detect structural variants that occur in patients with Pontine Tegmental Cap Dysplasia (PTCD), a rare neurological disorder

Undergraduate Research Assistant

Jun 2015-Mar 2016

Mentors: Dr. Julio Facelli, Ph.D. & Dr. Karen Eilbeck, Ph.D.; *Departments:* Biomedical Informatics & Genetics

University of Utah School of Medicine, Salt Lake City, UT

Scope: Investigated molecular descriptors of compounds that may decrease levels ATXN2 as a potential therapeutic for Spinocerebellar Ataxia Type 2 (SCA2); Mapped disease terms to the Human Phenotype Ontology (HPO); Redesigned a version of PhenoTips for clinical data entry

Undergraduate Research Assistant

Sep 2013-Sep 2014

Mentor: Dr. Patricia Jennings, Ph.D.; *Departments:* Chemistry & Biochemistry

University of California, San Diego, La Jolla, CA

Scope: Isolated MitoNEET protein & analyzed iron-sulfur clusters that regulate oxidative capacity in outer mitochondrial membranes

PEER-REVIEWED PUBLICATIONS

01. Imami A.S., **Sahay S.**, Creeden J.F.; drugfindR: Investigate iLINCS for candidate repurposable drugs. Bioconductor open-source software for bioinformatics (2026). <https://doi.org/10.18129/B9.bioc.drugfindR>.
02. Ryan W.G., **Sahay S.**, Vergis J., Weistuch C., Meller J., McCullumsmith R.E.; Pathway Analysis Interpretation in the Multi-Omic Era. BioTech 2025, 14(3), 58; [https://doi.org/10.3390-biotech14030058](https://doi.org/10.3390/biotech14030058).
03. Siciliano B., Henkel N.D., Ryan W.G., Imami A.S., Vergis J.M., Xu C., Arvay T.O., **Sahay S.**, Pulvender P., Hamoud A., Hales C., McCullumsmith R.E., Wen Z.; Pro-inflammatory transcriptomic and kinomic alterations in astrocytes derived from patients with familial Alzheimer's Disease. Brain, Behavior, and Immunity. 2025; Volume 47, 101044. <https://doi.org/10.1016/j.bbih.2025.101044>.
04. **Sahay S.**, Wen, J., Scoles D.R., Simeonov A., Jadhav A., Pulst S.M., Facelli J.C., Jones D.E.; Identifying Molecular Properties of Ataxin-2 Inhibitors for Spinocerebellar Ataxia Type 2

- utilizing High-Throughput Screening and Machine Learning. *Biology*. 2025; 522 (14). <https://doi.org/10.3390/biology14050522>.
05. **Sahay S.**, Rami Reddy M.V.S.R., Lennox C., Wolinsky E., McCullumsmith R.E., Singh T.; Harnessing Neuroimaging-Guided Transcranial Magnetic Stimulation for precision therapy in Substance Use Disorders. *Molecular Psychiatry*. 2025. <https://doi.org/10.1038/s41380-025-03024-x>.
 06. Lee C.-M., Nguyen J., Pope B., Imami A.S., Ryan W.G., **Sahay S.**, Mathis V., Pulvender P., Eby H., Arvay T., Alganem K., Wegmen-Points L., McCullumsmith R.E., Yuan L.; Functional kinome profiling reveals brain protein kinases signaling pathways and gene networks altered by acute voluntary exercise in rats. *PLoS ONE* 20(4): e0321596. <https://doi.org/10.1371/journal.pone.0321596>.
 07. Brenner M.J. *, **Sahay S.** *, Silveira R.M., Moser C., Morrison M.E., Zeitler N.K., Yang C.J., Colandrea M.T., McElroy K., Pandian V., (* **co-first authors**); Addressing Education and Care Gaps in Tracheostomy Management: Insights from a Multi-Stakeholder Global Survey. *Tracheostomy Journal*. 2025;2(1):15-28. <https://doi.org/10.62905/001c.129226>.
 08. **Sahay S.**, Lundh A.E., Sirole R.P., McCullumsmith R.E., O'Donovan S.M.; Purinergic System Transcript Changes in the Dorsolateral Prefrontal Cortex in Suicide and Major Depressive Disorder. *International Journal of Molecular Sciences*. 2025; 26(5): 1826. <https://doi.org/10.3390/ijms26051826>.
 09. Devine E., Imami A.S., Eby H., **Sahay S.**, Hamoud A., Golchin H., Ryan W., Shedroff E.A., Arvay T., Joyce A.W., Asah S.M., Consuelo W.B., O'Donovan S.M., McCullumsmith R.E.; Neuronal alterations in AKT isotype expression in schizophrenia. *Molecular Psychiatry*. 2024; 1-12. <https://doi.org/10.1038/s41380-024-02770-8>
 10. Hamoud A., Alganem K., Hanna S., Morran M., Henkel N., Imami A., Ryan W., **Sahay S.**, Pulvender P., Kunch A., Arvay T., Meller J., Shukla R., O'Donovan S., McCullumsmith R.; Illuminating the Dark Kinome: Utilizing Multiplex Peptide Activity Arrays to Functionally Annotate Understudied Kinases. *Cell Communication and Signaling*. 2024; 22(1):501. <https://doi.org/10.1186/s12964-024-01868-4>
 11. **Sahay S.**, Devine E., Vargas C.A.V., McCullumsmith R.E., O'Donovan S.M.; Adenosine Metabolism Pathway Alterations in Frontal Cortical Neurons in Schizophrenia. *Cells*. 2024; 13(19):1657. <https://doi.org/10.3390/cells13191657>
 12. **Sahay S.**, Vargas C.A.V., Hoar S.C., Choi J., O'Donovan S.M.; The Effects of the COVID-19 Pandemic on Mental Health among College Students in the United States. *Translation JMS*. 2024; 12(5):15. <https://doi.org/10.46570/utjms.vol12-2024-1280>
 13. **Sahay S.**, Pulvender P.G., Rami Reddy M.V.S.R., McCullumsmith R.E., O'Donovan S.M.; Metabolic insights into Neuropsychiatric Illnesses and Ketogenic Therapies: A Transcriptomic View. *International Journal of Molecular Sciences*. 2024; 25(15): 8266. <https://doi.org/10.3390/ijms25158266>
 14. **Sahay S.**, Hamoud A., Osman M., Pulvender P., McCullumsmith R.; Expression of WNT Signaling Genes in the Dorsolateral Prefrontal Cortex in Schizophrenia. *Brain Sciences*. 2024; 14(7):13. <https://doi.org/10.3390/brainsci14070649>
 15. **Sahay S.**, Devine E., McCullumsmith R.E., O'Donovan S.M.; Adenosine Receptor mRNA Expression in Frontal Cortical Neurons in Schizophrenia. *Cells*. 2024; 13(1):32. <https://doi.org/10.3390/cells13010032>
 16. **Sahay S.**, Henkel N.D., Vargas C.A.V., McCullumsmith R.E., O'Donovan S.M.; Activity of Protein Kinase A in the Frontal Cortex in Schizophrenia. *Brain Sciences*. 2024; 14(1):13. <https://doi.org/10.3390/brainsci14010013>
 17. Alganem K.*, Imami A.S.*, **Sahay S.***, Eby H., Arvay T.O., Abel M., Zhang X., McIntyre, W.B., Lee J., Au-Yeung C., Asgariroozbehani R., Panda R., Shukla R., O'Donovan S., Hahn M., Meller J., McCullumsmith R., (* **co-first authors**); Kaleidoscope: A Bioinformatics Web Application for

In Silico Exploration of Omics Data. *Journal of Bioinformatics and System Biology*. 2023; 6(4):12. <https://doi.org/10.26502/jbsb.5107068>

BOOK CHAPTERS

01. Eby H.*, Imami A.S.*, Ryan W.G.*, **Sahay S.**, Vergis J., Nguyen J.H., Arvay T.O., Pulvender P., Henkel N.D., McCullumsmith R.E.; What is the Best Evidence for Bioenergetic Abnormalities in Severe Mental Illness? 2025, Metabolic Neuropsychiatry. Online ISBN: 978-3-032-05630-6. https://doi.org/10.1007/978-3-032-05630-6_5.

ORAL PRESENTATIONS

01. **Sahay S.**, Tapescu I.; “Version 9 Staging for Salivary Gland Carcinoma” University of Michigan Medical School, Ann Arbor, MI, Jul 2026.
02. **Sahay S.** Bhatt A., Carpenter M., Miller R.; “Reactive Thrombocytosis in Iron Deficiency Anemia” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Apr 2026.
03. **Sahay S.**, Shabpiray H., “Clinical Management of Acute Pancreatitis” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Nov 2025.
04. **Sahay S.**, Kelly, V., “Differentiating Malingering vs. Conversion Disorder” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Oct 2025.
05. **Sahay S.**, O’Donovan S.M., McCullumsmith R.E., “Metabolic Perturbations in Severe Mental Illnesses / Doctoral Defense” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Mar 2025.
06. **Sahay S.**, McCullumsmith R.E., “Examining Purinergic Dysfunction in the Brain” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Jan 2025.
07. **Sahay S.**, “Polished, Persuasive, Professional: Conference Presentation Success” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Nov 2024.
08. **Sahay S.**, Giovannucci D., “Metamodulation of Glutamate Exocytosis” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Feb 2024.
09. **Sahay S.**, Potter D., McCullumsmith R.E., “Bioenergetic Dysfunction in Schizophrenia: Novel Findings & Treatment Approaches” CURESZ Foundation. The University of Cincinnati, Cincinnati, OH, Feb 2024.
10. **Sahay S.**, McCullumsmith R.E., O’Donovan S.M., “Purinergic System Perturbations in Schizophrenia” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Feb 2024.
11. **Sahay S.**, O’Donovan S.M., McCullumsmith R.E., “Metabolic Dysregulation across Neuropsychiatric Illnesses” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Jul 2023.
12. **Sahay S.**, McCullumsmith R.E., “Pathophysiology of Subarachnoid Hemorrhages” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Jun 2023.
13. **Sahay S.**, McCullumsmith R.E., O’Donovan S.M. “Gender Influencing Student Experiences in M.D.-Ph.D. Training” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Mar 2023.
14. **Sahay S.**, Giovannucci D., “Effects of beta-hydroxybutyrate on MK-801 induced schizophrenia-like behavior in mice” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Feb 2023.
15. **Sahay S.**, Hamoud A., McCullumsmith R.E., “Data Fabrication & Scientific Misconduct in Alzheimer’s Research” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Sep 2022.

16. **Sahay S.**, De La Serna I., “*Environmental Toxins: Ending the use of Agent Orange*” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Aug 2022.
17. **Sahay S.**, Jaume J., “*Gastrointestinal Pathophysiology: Acute Cholecystitis*” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Nov 2021.
18. **Sahay S.**, Sinclair D., Lundh A.E., Rethman J.B., Kakrecha O., O’Donovan S.M., “*The Neurobiology of MDD.*” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Jun 2021.
19. **Sahay S.**, McCullumsmith R.E., O’Donovan S.M., “*Gaining Insight into P2X Receptors & NLRP3 Inflammasome in SCZ Patients*” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Jun 2021.
20. **Sahay S.**, Henkel N.D., Kandimalla P., McCullumsmith R.E., “*Optimizing Protein Kinase A Activity Assay & H-89 Inhibitor*” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, Jun 2021.
21. **Sahay S.**, Alnafisah R.S., Wu X., Sinclair D., Kakrecha O., Lundh A.E., Bates E.A., O’Donovan S.M., “*The pH Meter & Spectrophotometer*” The University of Toledo College of Medicine & Life Sciences, Toledo, OH, May 2021.
22. **Sahay S.**, Jones D.E., Wen J, Facelli J.C., “*SCA2 Drug Repurposing*” University of Utah School of Medicine, Salt Lake City, UT, Aug 2015.

ABSTRACTS & CONFERENCE PROCEEDINGS

01. **Sahay S.***, Brenner M.J *, Silveira R.M.*, Moser C., Morrison M.E., Zeitler N.K., Yang C.J., Colandrea M.T., McElroy K., Pandian V.; Global Disparities in Tracheostomy Care Delivery and Education. *RUSH Otolaryngology Global Health Symposium*. Chicago, IL. May 2026.
02. Imami A.S., **Sahay S.**, Hamoud A., McCullumsmith R.E., Sex-Based Variations and Metabolic Patterns in Mental Health Disorders: Findings from MIMIC-IV. *Society of Biological Psychiatry Annual Meeting*. Toronto, ON. Apr 2025.
03. **Sahay S.**, Pulvender P., Rami Reddy M.V.S.R., McCullumsmith R.E., O’Donovan S.M.; Molecular Signatures of Metabolic Stress and Ketogenic Therapies in Severe Mental Illness. *Society of Biological Psychiatry Annual Meeting*. Toronto, ON. Apr 2025.
04. **Sahay S.**, Rami Reddy M.V.S.R., McCullumsmith R.E.; Transcranial Magnetic Stimulation for Substance Use Disorders. *Neuroscience & Psychiatry Symposium*. Toledo, OH. Feb 2025.
05. **Sahay S.**, Pulvender P., Rami Reddy M.V.S.R., McCullumsmith R.E., O’Donovan S.M.; Metabolic Insights into Neuropsychiatric Illnesses & Ketogenic Therapies: A Transcriptomic View. *Society for Neuroscience Annual Meeting*. Chicago, IL. Oct 2024.
06. **Sahay S.**, Imami A.S., O’Donovan S.M.; Sex-Specific Treatment of Major Depressive Disorder using Transcriptomic Profiling and LINCS analysis. *Great Lakes Bioinformatics Annual Meeting*. Pittsburgh, PA. May 2024. <https://doi.org/10.7490/f1000research.1119927.1>
07. **Sahay S.**, Imami A.S., Hamoud A., McCullumsmith R.E., O’Donovan S.M.; Towards timely Alzheimer’s diagnosis: a clinical data-driven approach to biomarker discovery. *American Academy of Neurology Annual Meeting*. Denver, CO. Apr 2024. *Neurology Journals*. 2024; 102(17). <https://doi.org/10.1212/WNL.00000000000205096>
08. **Sahay S.**, Devine E., McCullumsmith R.E., O’Donovan S.M.; Purinergic System Perturbations in Schizophrenia. *Graduate Research Annual Forum*. Toledo, OH. Mar 2024. *Translation*. 2024; 12(3). <https://doi.org/10.46570/utjms.vol12-2024-1223>
09. **Sahay S.**, Imami A.S., Hamoud A., McCullumsmith R.E., O’Donovan S.M.; Advancing the detection of early Alzheimer’s disease via clinical data mining. *Neuroscience & Psychiatry Symposium*. Toledo, OH. Jan 2024. <https://doi.org/10.46570/utjms-2024-1165>

10. **Sahay S.**, Dougherty D.M., Menon B.; Revamping the MMI: Enhancing Objectivity through Integration of AAMC Core Competencies. *AAMC Annual Meeting, Learn Serve, Lead*. Seattle, WA, Nov 2023.
11. **Sahay S.**, Devine E., McCullumsmith R.E., O'Donovan S.M.; Cell-Subtype changes in Adenosine Receptor Expression in Schizophrenia. *Ohio Physiological Society Annual Meeting*. Perrysburg, OH, Oct 2023.
12. Imami A.S., Hamoud, A., **Sahay S.**, McCullumsmith R.E.; Connecting the Dots: Understanding the Relationship Between Social Determinants & Schizophrenia. *Bioinformatics Symposium*. Toledo, OH, May 2023. <https://doi.org/10.46570/utjms-2025-1584>.
13. Imami A.S., Shedroff E.S., Joyce A.W., **Sahay S.**, Alganem K., Ryan W., Henkel N.D., McCullumsmith R.E.; Perturbation of Cell-Subtype Specific Active Kinome Networks in Schizophrenia. *Society for Neuroscience*. Chicago, IL, Nov 2021.
14. Kandimalla P., Henkel N.D., **Sahay S.**, McCullumsmith R.E.; Protein Kinase A Expression & Activity in Schizophrenia. *Society for Neuroscience*. Chicago, IL, Nov 2021.
15. **Sahay S.**, Moshiri N.; Variant Calling in *E. coli*. *Post-Baccalaureate Symposium*. Hayward, CA, May 2019.
16. Chong H., Hutchinson K., **Sahay S.**, Smith D.; A Platform for the Detection of HIV through Viral Load Testing in Tijuana, Mexico. *Undergraduate Research Symposium*. La Jolla, CA, Mar 2017.
17. Jones D.E., Wen J, **Sahay S.**, Scoles D.R., Dexheimer T., Hongmao S., Maloney D., Simeonov A., Jadhay A., Pulst S.M., Facelli J.C.; Identifying Candidate Ataxin-2 Inhibitors in High Throughput Screening Data using Molecular Descriptors. *AMIA Translational Summit*. San Francisco, CA, Jun 2015.

OTHER PUBLICATIONS

01. **Sahay S.**; Metabolic Perturbations in Severe Mental Illnesses. Doctoral Dissertation. Mar 2025. http://rave.ohiolink.edu/etdc/view?acc_num=mco1741284328366232
02. **Sahay S.**; The power struggle in our brain: Energy imbalance in mental health. The Toledo Blade. Dec 2024. <https://www.toledoblade.com/a-e/culture/2024/12/02/the-power-struggle-in-our-brain-energy-imbalance-in-mental-health/stories/20241202001>
03. **Sahay S.**; Health in the co-RONA 20s. Way To Goal. Jan 2021. <https://waytogoal.com/health-in-the-co-rona-20s/>

MANUSCRIPTS IN PREPARATION

01. **Sahay S.**, Kao D., Batey K., Van Horn A.; Occult Migrated Esophageal Foreign Body Presenting as a Pediatric Deep Neck Mass. *In preparation*.
02. Labilloy G., Funk A.J., **Sahay S.**, Reigle J., Alnafisah R., Golchin H., Heaven M.R., Roberts R.C., Shamsaei B., Greis K.D., Meller J., McCullumsmith R.E.; Region-Specific PSD-95 Interactomes Contribute to Functional Diversity of Excitatory Synapses in Human Brain. *Under review*.
03. **Sahay S.**, McCullumsmith R.E.; Bioenergetic Dysfunction in Neuropsychiatric Illnesses: The Five Fates of Glucose. *In preparation*.
04. Imami A.S.*, **Sahay S.*** (* **co-first authors**); drugfindR: Investigate iLINCS for candidate repurposable drugs | Vignette. Bioconductor. *In preparation*.
05. Imami A.S.*, **Sahay S.***, O'Donovan S.M. (* **co-first authors**); DrugfindR: Application Note. BMC Bioinformatics. *In preparation*.

JOURNAL REVIEW BOARDS

Translation Editorial Board Member & Student Reviewer

Jun 2022-Present

Translation: The University of Toledo Journal of Medical Sciences

Scope: Review 4-6 manuscripts per year for this academic, peer-reviewed journal

ADMINISTRATIVE & LEADERSHIP ACTIVITIES

Department of Neurosciences & Psychiatry I&O Committee Member Sep 2022-Present

Scope: Provide advice, integrate initiatives, & ensure accountability on Inclusion & Opportunities (I &O)

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

MD Admissions Committee Member Oct 2021-Present

Primary Review Subcommittee: Conduct primary evaluations & vote on inviting prospective medical students for an interview; Conduct interviews for prospective medical students

Final Review Subcommittee: Conduct final evaluations & vote on prospective medical students; Conduct interviews for prospective medical students

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

College of Biomedical Graduate Studies Executive Board Member Aug 2022-May 2024

Scope: Acted as a contact point & liaison for M.D./Ph.D. students; organized career fairs, research meetings, professional development workshops, & team-building events

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Multiple Mini Interview (MMI) Committee: Student Lead Jul 2022-Oct 2023

Scope: Lead & work with a student group to enhance the quality of the MMI interview prompts & overall process for incoming medical students using a mission-oriented approach

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Vice President of Otolaryngology Board Aug 2020-Aug 2022

Scope: Aided students interested in otolaryngology match with physicians in the community for observership programs; Organize events such as information panels, suturing events, & various learning opportunities for interested medical students

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

TEACHING EXPERIENCE

Private 1:1 Tutor Mar 2019-May 2024

Scope: Specialize in ELA subject area, standardized test preparation, study organization skills & coaching throughout the college admissions process

Private 1:1 "Expert" Tutor Sep 2018-Sep 2022

Tutor Corps, San Francisco, CA

Scope: Specialized in ELA & STEM subject areas, & standardized test preparation; Worked with students who had learning differences or came from underserved communities

ESL Volunteer Tutor Jan 2019-Jul 2020

Pleasanton Reads, Pleasanton, CA

Scope: Built relationships with underserved residents in the community to improve written & conversational English literacy skills; Recruited volunteers to expand the program & decreased the student: instructor match time

Instructor Sep 2019-Jul 2020

Pacific Preparatory School, San Francisco, CA

Scope: Taught & provided behavioral support for students with learning differences by leveraging Common Core education standards to develop novel curriculum for ELA & STEM subject areas

Graduate Teaching Assistant

Sep 2017-Aug 2018

California State University, East Bay, Hayward, CA

*Scope: Helped students develop a depth of understanding of course material in graduate level science courses, such as epidemiology & advanced biochemistry, by holding office hours & supplemental teaching sessions***Volunteer Tutor**

Jan 2015-Nov 2017

Reality Changers, San Diego, CA

*Scope: Mentored over 25 students from low socioeconomic backgrounds in ELA & STEM subject areas who aspired & achieved their goal to become first-generation college students***MENTORSHIP & OUTREACH**

Mentoring Role: Research Advisor in Basic Sciences

Jan 2024-Apr 2025

Mentee: Madhu Rami Reddy; Last known position: Medical student

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Mentoring Role: Research Advisor in Basic Sciences

Jan 2024-Feb 2025

Mentee: Roshan Sirole; Last known position: Undergraduate student

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Mentoring Role: Research Advisor in Basic Sciences

Dec 2022-Oct 2024

Mentee: Christina Flora-Annabelle Vargas; Last known position: Undergraduate student

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Mentoring Role: Research Advisor in Basic Sciences

Jun 2023-Jul 2024

Mentee: Priyanka Pulvender; Last known position: Medical student

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Mentoring Role: Research Advisor in Basic Sciences

May 2024-Aug 2024

Mentee: Ojal Kakrecha; Last known position: Medical student

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Mentoring Role: Research Lab Camp Mentor

Jul 2023, Jul 2024

Department of Neurosciences High School Summer Lab Camp

Scope: Mentored students from Toledo regional high schools in wet laboratory techniques & bioinformatics tools to pique their interest in STEM & research

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Mentoring Role: Research Advisor in Basic Sciences

May 2023-Dec 2023

Mentees: Sophia Hoar; Jae Choi; Last known position: Undergraduate students

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Mentoring Role: AMSA Pre-Medical Student Mentor

May 2021-May 2023

American Medical Student Association (AMSA) x Ambassadors Mentorship Program

Mentee: Sophia Durham; Last known position: Undergraduate Student

Jul 2022-May 2023

Mentee: Joseph Raimonde; Last known position: Medical Student

May 2021-Jul 2022

Mentee: Rita Kamoua; Last known position: Medical student

May 2021-Jul 2022

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Mentoring Role: Research Advisor in Basic Sciences

Aug 2022-Jan 2023

Mentees: Insiya Rasheed, Annika Padhye; Last known position: Undergraduate students

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Mentoring Role: REACH Program Volunteer

Jun 2022

The Research Experience & Awareness of Clinical Health (REACH) Program

Scope: Served as a volunteer for this nationally recognized program to expose under-represented high school students to clinical research & experiences; Held information panels, discussion sessions, & helped students with research presentations

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Mentoring Role: Research Advisor in Basic Sciences

May 2021-Aug 2021

Mentees: Anna Lundh, Jake Rethman, Ojal Kakrecha; Last known position: Medical students

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

SERVICE & OUTREACH

Medical Student Volunteer, ENT Free Clinic

May 2026-Present

Hope Clinic, Ann Arbor, MI

Scope: Perform interviews & physical exams on patients under the supervision of otolaryngologists; Provide audiograms, endoscopies, ear procedures, biopsies, excisions of cancerous lesions, & examinations of serious eye conditions for underserved, uninsured patients in the Ypsilanti area

Medical Student Volunteer, Free Clinic

Jul 2022-Present

Community Care Clinics, Toledo, OH

Scope: Perform interviews & physical exams on patients under the supervision of physicians & mid-level providers; Provide adult vision screenings, access to high-quality comprehensive healthcare, & free medications for underserved, uninsured patients in Northwest Ohio through the largest free student-run clinic in the country

Medical Student Volunteer

Jan 2021-Feb 2022

ProMedica, Toledo, OH

Scope: Performed basic components of physical exams on patients under the supervision of physicians in the fields of family medicine, neurology, & psychiatry via the Integrated Clinical Experience Program through The University of Toledo & ProMedica Partnership

Medical Student Volunteer

Aug 2020-May 2021

Community Care Vaccine Clinics, Toledo, OH

Scope: Distributed COVID-19 vaccines to members of the Greater Toledo community

Hospital Volunteer

Jan 2015-Dec 2017

Scripps Ximed Surgery Center, La Jolla, CA

Scope: Designed a data-driven solution utilizing hospital layout & patient traffic to prevent overcrowding in the surgery center; Assisted the post-operation team to ensure patients were stable before being released from the hospital; Trained & onboarded volunteers on the workflow of the surgery center from medical charting to post-operation services

WORK EXPERIENCE

Healthcare Data Research Analyst

May 2020-Aug 2021

Way To Goal, LLC, Phoenix, AZ

Scope: Conducted research studies that offered marketing insights to biomedical companies looking to increase market share of their products; Collected, & analyzed data in a variety of therapeutic areas across many stages of the product development cycle; Provided actionable recommendations for major Fortune 500 companies

Medical Assistant

Aug 2017-Apr 2018

Allied Pain & Spine Institute, San Jose, CA

Scope: Redesigned clinical training guidelines to increase patient-provider appointment efficiency; Managed front & back-office tasks: instructing patients about high-risk procedures, overseeing physicians' schedules, & acting as a liaison with insurance companies & clinics

Healthcare Marketing Intern

Jan 2014-Jun 2015

BioVision Incorporated, Milpitas, CA

Scope: Provided research laboratories & local businesses in the San Diego area with information about BioVision brand technology & products via informational seminars & tabletop presentations

GRANTS

Short-term Fellowship Research Training Grant (T15-LM007124-18)

Jun 2015

National Library of Medicine

Scope: In silico investigation of chemical properties of a potential Spinocerebellar Ataxia Type II therapeutic

AWARDS & HONORS

M.D./Ph.D. Dean's Scholarship

Jul 2021-Present

The University of Toledo College of Medicine & Life Sciences, Toledo, OH

Second Place Poster Presentation

Oct 2023

The Annual Ohio Physiological Society Meeting, Perrysburg, OH

First Place Poster Presentation

May 2019

Genetics Subdivision at Post Baccalaureate Research Symposium
California State University, East Bay, Hayward, CA

Merit Scholarship

Sep 2017-Aug 2018

California State University, East Bay, Hayward, CA

First Place: Open Viral Load Model

Jul 2017

\$25K Grand Prize at Global Grand Challenges Summit: Model & Poster Competition
USA, UK, & Chinese National Academies of Engineering, Washington, D.C.

Provost's Honors

Sep 2016-Jun 2017

University of California, San Diego, La Jolla, CA

PROFESSIONAL ASSOCIATIONS

American Association for the Advancement of Science – Student Member

The American Physician Scientists Association – Student Member

The American Medical Association – Student Member

College of Biomedical Graduate Studies – Past Board Member, Current Student Member

Graduate Student Association – Student Member

University of Michigan Medical School ENT Interest Group – Student Member

UTCOMLS ENT Interest Group – Past Board Member, Current Student Member

UTCOMLS Neurology Interest Group – Past Student Member

TECHNICAL SKILLS

Statistical / Machine Learning Tools: R, GraphPad, Statistica, WEKA

Bioinformatics Tools: Kaleidoscope, Galaxy, NCBI GenBank & BLAST, OMIM, Primer3, ApE, Cap3, Sequence Assembly Program, ExPASy, InterPro, Clustal Omega, T-Coffee, Jalview, MEGA, Genevestigator, UniProt

Laboratory Techniques: Cryostat sectioning, RNA isolation, cDNA synthesis, RT-qPCR, protein quantification assays, protein activity assays, laser microdissections, chromatography, immunohistochemistry, gel electrophoresis, enzyme restriction digestions, immunoblotting

LANGUAGES

English, Hindi, Urdu